

Causal Effect of Private Insurance on Flu Vaccination

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1 Research Question and Data

1.1 Research Question

This project studies the causal effect of private health insurance coverage on the probability of receiving a flu vaccination.

The treatment variable is:

$$A = \text{Private insurance coverage}$$

where:

$$A = \begin{cases} 1 & \text{Any private insurance} \\ 0 & \text{Uninsured} \end{cases}$$

The outcome variable is:

$$Y = \begin{cases} 1 & \text{Received flu shot during the past year} \\ 0 & \text{Did not received flu shot during the past year} \end{cases}$$

Our target estimand is the average treatment effect (ATE):

$$\tau = E[Y(1) - Y(0)]$$

where $Y(1)$ and $Y(0)$ denote the potential flu-vaccination outcomes under private insurance and no insurance respectively.

We restrict the analysis to adults because insurance enrollment and vaccination decisions for children are typically made by parents or guardians rather than by the children themselves. As a result, the causal relationship between insurance coverage and flu vaccination among children reflects household-level decision making instead of individual behavioral choices. In addition, pediatric vaccination schedules and eligibility for public insurance programs differ substantially from those of adults. To reduce this source of heterogeneity, the analysis focuses on adults only.

Furthermore, the treatment definition is restricted to privately insured versus uninsured individuals. Public insurance programs often involve different eligibility rules, healthcare access mechanisms, and demographic compositions, which could introduce additional heterogeneity into the treatment effect estimation.

1.2 Data Source

We use the 2018 Medical Expenditure Panel Survey (MEPS) Full-Year Consolidated File. The raw dataset contains 30,461 observations and 1,501 variables.

The insurance variable used in the analysis is INSCOV18, whose official MEPS coding is:

Code	Meaning
1	Any private insurance
2	Public insurance only
3	Uninsured

Table 1: MEPS insurance coding

After restricting the sample to private insurance versus uninsured individuals only, the final analytic dataset contains 13,612 observations with 11 variables.

Variable	Count
Privately insured	11,992
Uninsured	1,620
Received flu shot	5,638
Did not receive flu shot	7,974

Table 2: Treatment and outcome distributions

The observed covariates used for adjustment are:

$$Z = \{\text{Age, Race, Poverty Status, Region,} \\ \text{Physical Health, Mental Health, Education}\}$$

2 Causal Model

2.1 DAG Specification

Figure 1 presents the assumed causal directed acyclic graph (DAG) for the relationship between private insurance and flu vaccination.

Full DAG: Insurance and Flu Vaccination

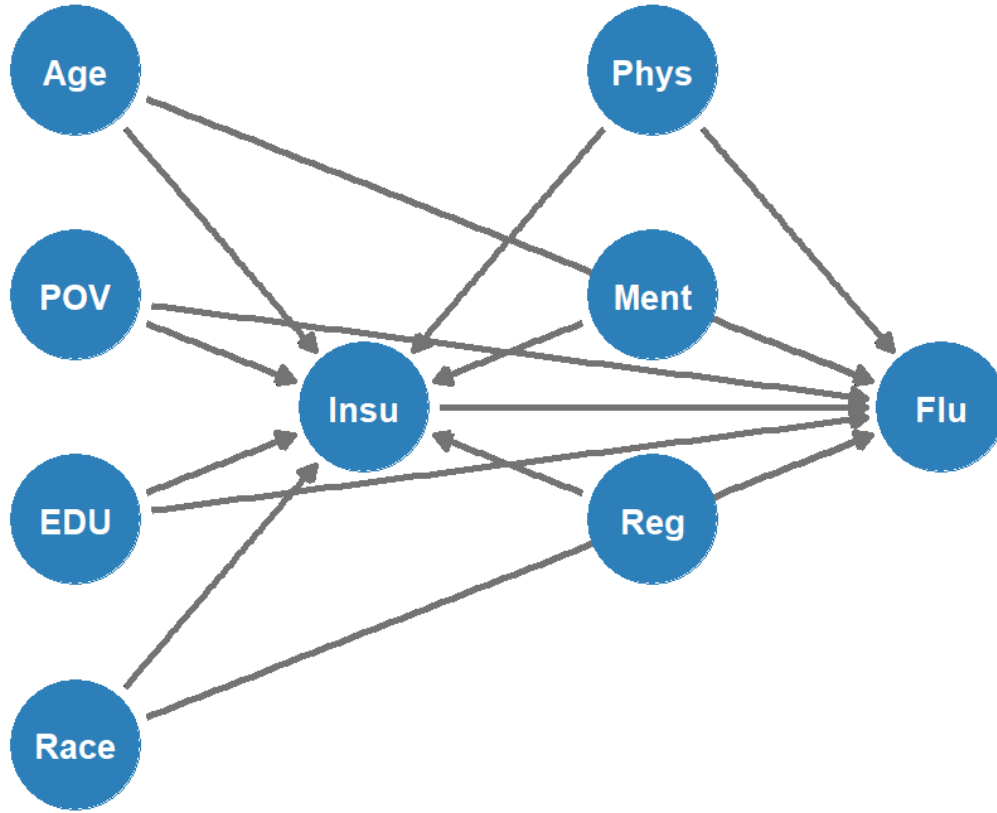


Figure 1: Causal DAG for private insurance and flu vaccination

The DAG assumes that private insurance coverage directly affects flu vaccination behavior:

$$\text{Insurance} \rightarrow \text{FluShot}$$

Several observed variables are modeled as confounders because they plausibly affect both insurance status and vaccination probability:

- Age influences insurance eligibility and preventive care usage.
- Poverty and education affect access and willingness to insurance and healthcare utilization.
- Race may affect healthcare access and vaccination behavior.
- Physical and mental health conditions may increase both insurance enrollment and vaccine uptake.
- Geographic region captures differences in healthcare systems and vaccination campaigns.

The DAG follows the standard back-door framework.

2.2 Potential Unmeasured Confounding

Although many observed confounders are included, the DAG may still omit latent variables such as:

- Health awareness
- Risk aversion
- Preventive-care attitudes
- Trust in healthcare systems

These unobserved variables may influence both insurance enrollment and vaccination behavior simultaneously, producing residual confounding. This concern motivates the later sensitivity analysis using E-values.

3 Identification Strategy

The identification strategy relies on the back-door adjustment criterion. Based on the assumed DAG, the observed covariates block all non-causal back-door paths from private insurance to flu vaccination.

Using `dagitty`, the minimal adjustment set is:

{Age, Education, Mental Health, Poverty, Physical Health, Race, Region}.

Equivalently, in the variables used in the analysis, the adjustment set is:

$Z = \{\text{age, education_years, mental_health,}$
 $\text{poverty_category, physical_health,}$
 $\text{race_ethnicity, region}\}.$

Under consistency, conditional exchangeability, and positivity, the causal effect is identified by:

$$E[Y \mid do(A = a)] = \sum_z E[Y \mid A = a, Z = z] P(Z = z).$$

Therefore, the average treatment effect is:

$$\tau = E[Y(1) - Y(0)] = \sum_z [E(Y \mid A = 1, Z = z) - E(Y \mid A = 0, Z = z)] P(Z = z).$$

Here, $A = 1$ represents private insurance, $A = 0$ represents being uninsured, and $Y = 1$ represents receiving a flu shot.

4 Estimation

Using the adjustment set identified in Section 3, we estimate the average treatment effect (ATE) of private insurance coverage on flu vaccination using three commonly used causal inference estimators: outcome regression (OR), inverse probability weighting (IPW), and augmented inverse probability weighting (AIPW). All methods adjust for age, race, education, poverty status, region, physical health, and mental health. We additionally construct bootstrap confidence intervals to quantify statistical uncertainty.

4.1 Outcome Regression

Outcome regression estimates the conditional probability of flu vaccination given treatment status and covariates. The validity of outcome regression depends on correct specification of the outcome model and the assumptions of consistency, conditional exchangeability, and positivity. We fit a logistic regression model for flu vaccination,

$$P(Y = 1 \mid A, Z),$$

where Y denotes flu vaccination status, A denotes private insurance coverage, and Z represents the adjustment variables.

After fitting the model, we estimated the counterfactual vaccination probability under treatment by setting every individual's insurance status to $A = 1$ and predicting the corresponding outcome probability. Similarly, we estimated the counterfactual vaccination probability under control by setting $A = 0$ for all individuals. The average treatment effect was estimated as

$$\hat{\tau}_{OR} = \frac{1}{n} \sum_{i=1}^n \hat{m}(1, Z_i) - \frac{1}{n} \sum_{i=1}^n \hat{m}(0, Z_i),$$

where $\hat{m}(a, Z)$ denotes the fitted outcome regression model.

4.2 Inverse Probability Weighting

Inverse probability weighting (IPW) uses the propensity score to construct a pseudo-population in which treated and untreated individuals have more similar distributions of observed covariates. Diagnostic checks indicated substantial improvement in covariate balance after weighting, though balance was not perfect for all variables. The validity of the IPW estimator depends on correct specification of the propensity score model together with the assumptions of consistency, conditional exchangeability, and positivity.

We first estimated the propensity score

$$e(Z) = P(A = 1 \mid Z)$$

using logistic regression with the identified adjustment variables.

We use the Hájek form of the inverse probability weighting estimator, which normalizes the inverse probability weights within each treatment group. The estimator of the average treatment effect is

$$\hat{\tau}_{IPW} = \frac{\sum_{i=1}^n \frac{A_i Y_i}{\hat{e}(Z_i)}}{\sum_{i=1}^n \frac{A_i}{\hat{e}(Z_i)}} - \frac{\sum_{i=1}^n \frac{(1 - A_i) Y_i}{1 - \hat{e}(Z_i)}}{\sum_{i=1}^n \frac{1 - A_i}{1 - \hat{e}(Z_i)}}.$$

Intuitively, observations that receive an unlikely treatment assignment given their covariates receive larger weights, which helps balance the treated and control groups. The Hájek estimator additionally normalizes these weights within treatment groups, often improving stability when some estimated propensity scores are close to 0 or 1.

4.3 Augmented Inverse Probability Weighting

Augmented inverse probability weighting combines outcome regression and propensity score weighting. The AIPW estimator is doubly robust, meaning it remains consistent if either the outcome model or the propensity score model is correctly specified.

The estimator is given by

$$\hat{\tau}_{AIPW} = \frac{1}{n} \sum_{i=1}^n \left[\hat{m}(1, Z_i) - \hat{m}(0, Z_i) + \frac{A_i(Y_i - \hat{m}(1, Z_i))}{\hat{e}(Z_i)} - \frac{(1 - A_i)(Y_i - \hat{m}(0, Z_i))}{1 - \hat{e}(Z_i)} \right].$$

Because AIPW combines two modeling strategies, it is often more robust than either outcome regression or IPW alone.

4.4 Bootstrap Confidence Intervals

To quantify uncertainty, we used a nonparametric bootstrap procedure with 500 bootstrap resamples. For each bootstrap sample, we re-estimated the treatment effect using outcome regression, IPW, and AIPW. We then constructed 95% confidence intervals using the empirical 2.5th and 97.5th percentiles of the bootstrap distribution.

Estimator	Estimate	95% CI Lower	95% CI Upper
Naive	0.318	0.299	0.337
Outcome Regression	0.220	0.193	0.249
Hájek IPW	0.281	0.240	0.316
AIPW	0.267	0.224	0.303

Table 3: Estimated average treatment effects of private insurance on flu vaccination.

Table 3 summarizes the estimated average treatment effects obtained using several estimation methods. Although the estimated magnitudes differ somewhat across methods, all adjusted estimators produce positive effects of similar order and lead to the same substantive conclusion. The consistency in direction across outcome regression, Hájek IPW, and AIPW provides some reassurance that the estimated positive effect is not driven by a particular modeling strategy.

5 Diagnostics

5.1 Positivity

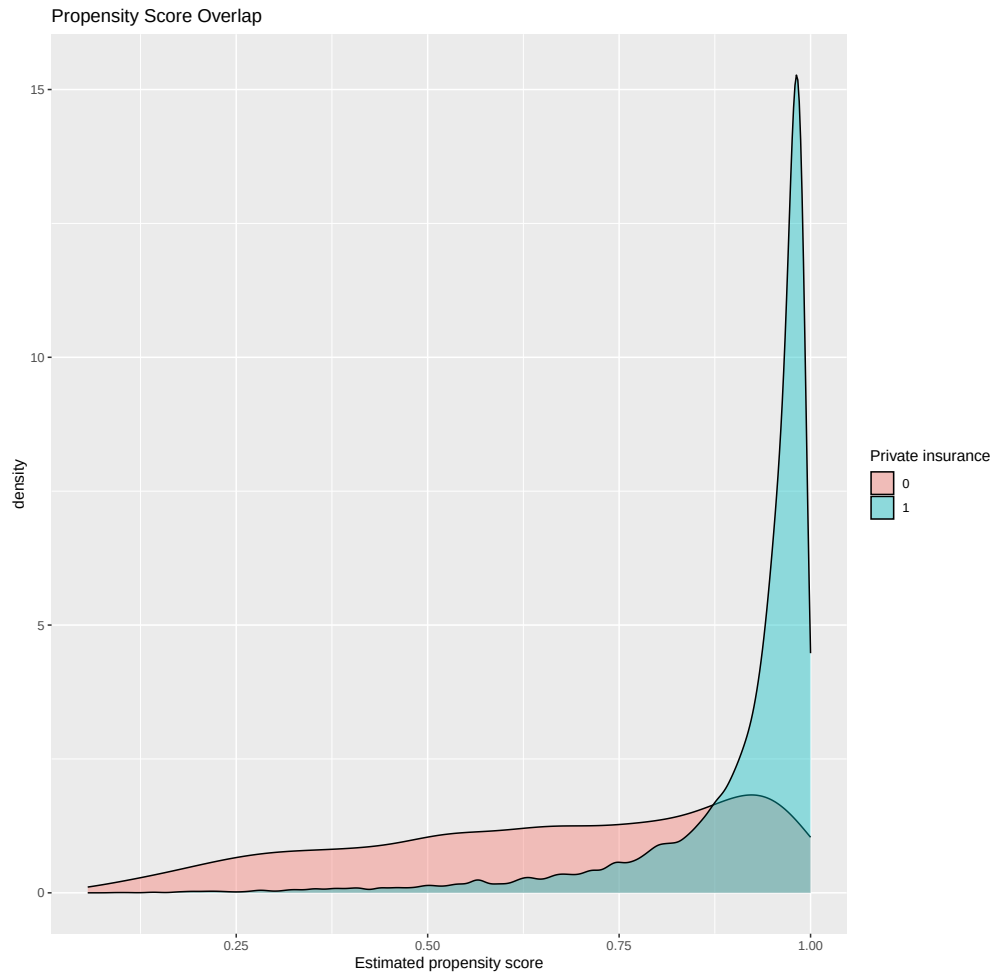


Figure 2: Distribution of estimated propensity scores for privately insured and uninsured individuals.

Although overlap exists across much of the observed support, privately insured individuals are heavily concentrated at propensity-score values close to one, indicating limited overlap in some regions of the covariate space. This suggests that the positivity assumption is broadly plausible but not ideal. Because inverse probability weighting can be sensitive to extreme propensity scores, the IPW estimates may be less stable in regions with limited overlap. For this reason, greater emphasis is placed on the outcome regression and AIPW estimates, which are generally more robust in settings with imperfect overlap.

5.2 Covariate Balance

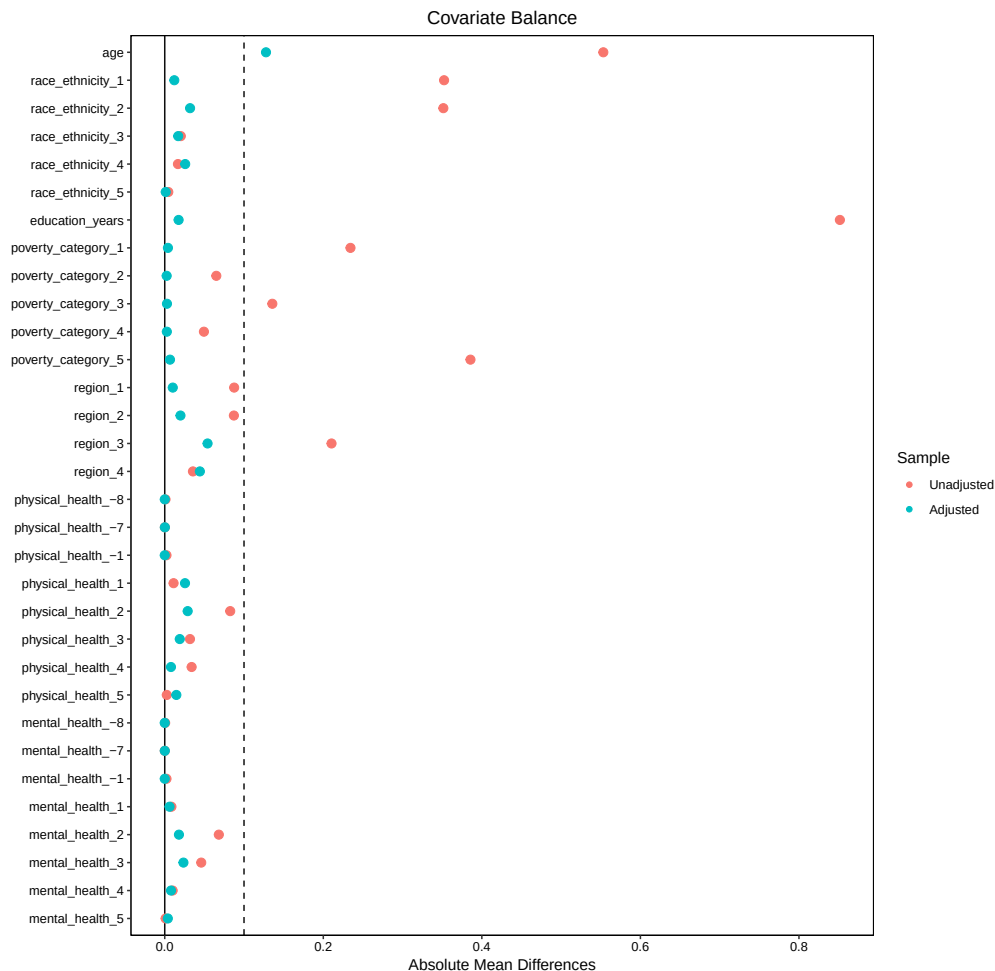


Figure 3: Covariate balance before and after inverse probability weighting.

Figure 3 presents the standardized mean differences before and after inverse probability weighting. Prior to weighting, several covariates exhibited noticeable imbalance, particularly age, education, poverty status, and race. After weighting, the standardized mean differences were substantially reduced, with most covariates falling below the commonly used threshold of 0.1. This indicates that the weighting procedure improved covariate balance between the treatment and control groups.

5.3 Sensitivity Analysis

Although we adjusted for a rich set of observed covariates, unmeasured confounding may still be present. Because flu vaccination is a binary outcome and our primary analysis is based on logistic outcome models, we assess sensitivity to unmeasured confounding using the E-value, which is naturally interpreted on the risk-ratio scale.

The standardized outcome regression estimated a risk ratio of 2.025 for flu vaccination comparing privately insured and uninsured individuals. The corresponding E-value was 3.47.

This implies that an unmeasured confounder would need to be associated with both private insurance coverage and flu vaccination by a risk ratio of at least 3.47 each, above and beyond the measured covariates, to completely explain away the observed association.

The E-value therefore suggests a moderate degree of robustness to unmeasured confounding. While residual confounding cannot be ruled out, any omitted confounder would need to have a relatively strong association with both insurance coverage and flu vaccination in order to fully account for the estimated effect.

Potential candidates include factors that are not available in the MEPS data, such as health awareness, preventive-care attitudes, risk aversion, and trust in healthcare systems. If such factors are strongly related to both insurance enrollment and vaccination behavior, some bias may remain. However, the E-value indicates that relatively substantial confounding would be required to eliminate the observed association entirely.

6 Results and Interpretation

6.1 Main Findings

Across all estimation strategies, private insurance coverage was associated with a higher probability of receiving a flu vaccination. The naive comparison estimated a 31.8 percentage-point increase in vaccination probability among privately insured individuals compared with uninsured individuals. After adjusting for observed confounders, the estimated effects were smaller but remained positive.

The outcome regression estimate was 0.220, while the Hájek IPW and AIPW estimates were 0.281 and 0.267, respectively. The reduction from the naive estimate suggests that part of the observed association is explained by differences in age, education, poverty status, race, region, and health status between privately insured and uninsured individuals.

Despite differences in magnitude, all adjusted estimators produced similar conclusions. Furthermore, the bootstrap confidence intervals excluded zero for all adjusted estimators, providing evidence that private insurance coverage is associated with an increased probability of receiving a flu vaccination.

6.2 Assumptions and Limitations

The validity of our causal estimates relies on the assumptions of consistency, conditional exchangeability, and positivity. Among these assumptions, conditional exchangeability is the most critical and also the most difficult to verify. Our identification strategy assumes that, after adjusting for age, education, poverty status, race, region, physical health, and mental health, no remaining confounders jointly influence both private insurance coverage and flu vaccination.

The positivity assumption appears broadly reasonable but is not perfectly satisfied. The propensity score overlap plot shows substantial overlap between treatment groups across much of the observed covariate space. However, privately insured individuals are heavily

concentrated at high propensity-score values, indicating limited overlap in some regions where private insurance coverage is nearly universal. As a result, causal estimates in those regions rely on fewer comparable uninsured individuals.

The diagnostic analyses provide additional support for the identification assumptions. Covariate balance improved substantially after weighting, with most standardized mean differences falling below conventional thresholds. In addition, the E-value analysis suggests that a relatively strong unmeasured confounder would be required to fully explain away the observed association.

Consequently, our analysis can conclude that private insurance coverage is associated with a higher probability of receiving a flu vaccination and that this conclusion is robust across several adjustment-based estimators. Under the assumed DAG and identification assumptions, the results are consistent with a positive causal effect of private insurance coverage on flu vaccination. However, the analysis cannot definitively rule out bias from unmeasured confounding, and the estimated causal effect should therefore be interpreted with appropriate caution.

7 Broader Context

One important causal inference method covered in the course that we did not use is instrumental variables (IV). IV methods are particularly valuable in observational studies when unmeasured confounding is suspected, because they can identify causal effects without requiring the strong assumption of conditional exchangeability. In principle, IV methods could be appealing in our setting since factors such as health awareness or preventive-care attitudes may remain unobserved.

However, finding a credible instrument for private insurance coverage is difficult. A valid instrument must be strongly associated with insurance status while affecting flu vaccination only through insurance coverage. In practice, many variables that influence insurance enrollment, such as healthcare access, employment conditions, or local policy environments, may also directly affect vaccination behavior, violating the exclusion restriction. Furthermore, the MEPS data used in this study do not contain an obvious candidate variable that plausibly satisfies the IV assumptions.

One possible extension of this analysis would be to incorporate causal discovery methods, such as PC or GES, to explore whether the assumed DAG is supported by the observed data. While domain knowledge remains essential for causal interpretation, causal discovery could provide an additional structural sensitivity analysis and help assess the robustness of the chosen adjustment set.